HW 1

1.1 Problem 14

a) If you have the flu, you miss the final exam

b) You pass the course if and only if you don't miss the final exam

c) If you miss the final exam, you don't pass the course

d) You have the flu or you miss the final exam or you pass the course

e) If you have the flu, you don't pass the course or if you miss the final exam you don't pass the course

f) If You have the flu and you miss the final exam or you don't miss the final exam and you pass the course

Problem 26

a) If I remember to send you the address, then you sent me an email message

b) If you were born in the U.S., then you are a citizen of this country

c) If you keep your textbook, then it will be a useful reference in your future courses

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d) If their goalie plays well, then The Red Wings will win the Stanley Cup

e) If you get the job, then you had the best credentials

f) If there is a storm, then the beach eroded

g) If you logged on to the server, then you had the valid password

h) If you begin your climb on time, then you will reach the summit

i) If you are among the 1st 100 customers tomorrow, then you will get a free icecream cone

Problem 34

a)

p	$\neg p$	$p \to \neg p$
T	F	F
F	T	T

b)

p	$\neg p$	$p \leftrightarrow \neg p$
T	F	F
F	T	F

c)

p	q	$p \vee q$	$p \oplus (p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	F

d)

p	q	$p \wedge q$	$p \vee q$	$(p \wedge q) \to (p \vee q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

e)

p	q	$\neg p$	$q \to \neg p$	$p \leftrightarrow q$	$(q \to \neg p) \leftrightarrow (p \leftrightarrow q)$
T	T	F	F	T	F
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T

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